

Plantim Icons **v4** — process report

Designer-grade rework of the full semantic icon set for web (Vue 3), iOS (SwiftUI) and Android (Vector Drawable / Compose).
Hand-authored geometry, four variants sharing one silhouette, six optical sizes on three authored grades, semantic color roles,
and mechanical + human gates.

registry 4.0.0 · v4Hash 573a185872b44f38b596f04ea43fd9241a433a2722d278ae1533bc80d82384b1

Why v4

The v3 experiment derived multicolor and multi-size variants from v2 programmatically. The audit (`reports/v4-v3-audit.md`) judged ~30% of the set unusable and catalogued the failure modes: fake solids (thicker outlines instead of fills, or fills that erase identity), wrong source glyphs mapped to semantic ids (a closed eye for `utility.eye`, an unlocked padlock for `utility.lock`, a question-mark bubble as the Chat tab), variant identity collisions, multicolor roles assigned by geometry heuristics, and blob anatomy across the hand-built garden/family/genus sets.

Method

- **App audit first.** 1,083 icon references across the Plantim web and iOS clients defined the tiers: 9 P0 tab/section identities, 177 P1 app-used icons, 103 P2 library icons (`reports/v4-app-icon-audit.md`).
- **Written design language.** 24-grid, 2 px round strokes, three authored optical grades (micro 16/20 @2.4 · base 24/32 @2.0 · display 48/72 @1.6), named layers with hand-assigned semantic color roles, separately authored solids with declared counters (`design-tokens/icons/v4/DESIGN.md`).
- **Mechanical gates.** Structural: hash integrity, path-walking bounds, stroke discipline, determinism. Raster: flood-fill counter verification (accidental islands fail) and outline ↔ solid silhouette IoU ≥ 0.90.
- **Human loop.** Every icon rendered to contact sheets (variants × sizes × light/dark; P0 additionally in iOS and Material 3 tab-bar mocks) and inspected zoomed; reworked until it reads. P0 icons got a minimum of two full passes (`reports/v4-worklog.md`).

Results

TIER	COUNT	AUTHORED GRADES
P0 — tab/section identity, animated	9	micro + base + display
P1 — used by the app	177	micro + base (display derived)
P2 — library	103	base (micro/display derived)
total	289 icons · 6936 SVGs	4 variants × 6 sizes

P0 case studies (v3 → v4)

BEFORE → AFTER	NOTES
  →   	<code>plant.growth</code> Gardens tab: v3's slashed leaf read as 'prohibited'; v4 draws a raised bed with two staggered plants — a collection, distinct from the single sprout at every size.
  →   	<code>plant.sprout</code> Plants tab: leaves opened so the outline keeps interior white at 24 px; solid drawn to the stroked footprint; ground/stem/leaves carry semantic color roles.
  →   	<code>utility.message.circle</code> Chat tab: v3 shipped a question-mark bubble. v4: round bubble with typing dots that survive the solid as counters.
  →   	<code>calendar.date</code> Rhythm tab: hand-drawn frame, rings and an emphatic date dot; the solid keeps a header slit + dot as counters instead of going black.

	<code>account.user</code> Profile: v3 multicolor erased the person (geometry-heuristic roles); v4 assigns roles by hand.
	<code>navigation.home</code> Today: closed-shell house whose silhouette holds; the door survives the solid as an open-bottom cutout.

Platform delivery

- **Web:** `@plantim/icons/v4` subpath — tree-shakable per-icon modules, CSS-variable token theming, the v2 entry byte-untouched and its size budgets unchanged.
- **iOS:** `PlantimIconV4` SwiftUI view — native Path rendering, grade selection by size, dynamic light/dark token colors, tab-selection animations honoring Reduce Motion.
- **Android:** `packages/plantim-icons-android` — tintable Vector Drawables (values-night palette) and Compose ImageVectors with all grades, xmllint-validated.
- **Motion:** P0 selection animations compiled to Lottie JSON from declarative specs (220–260 ms, transform/opacity/trim subset).